

Colin Balfour

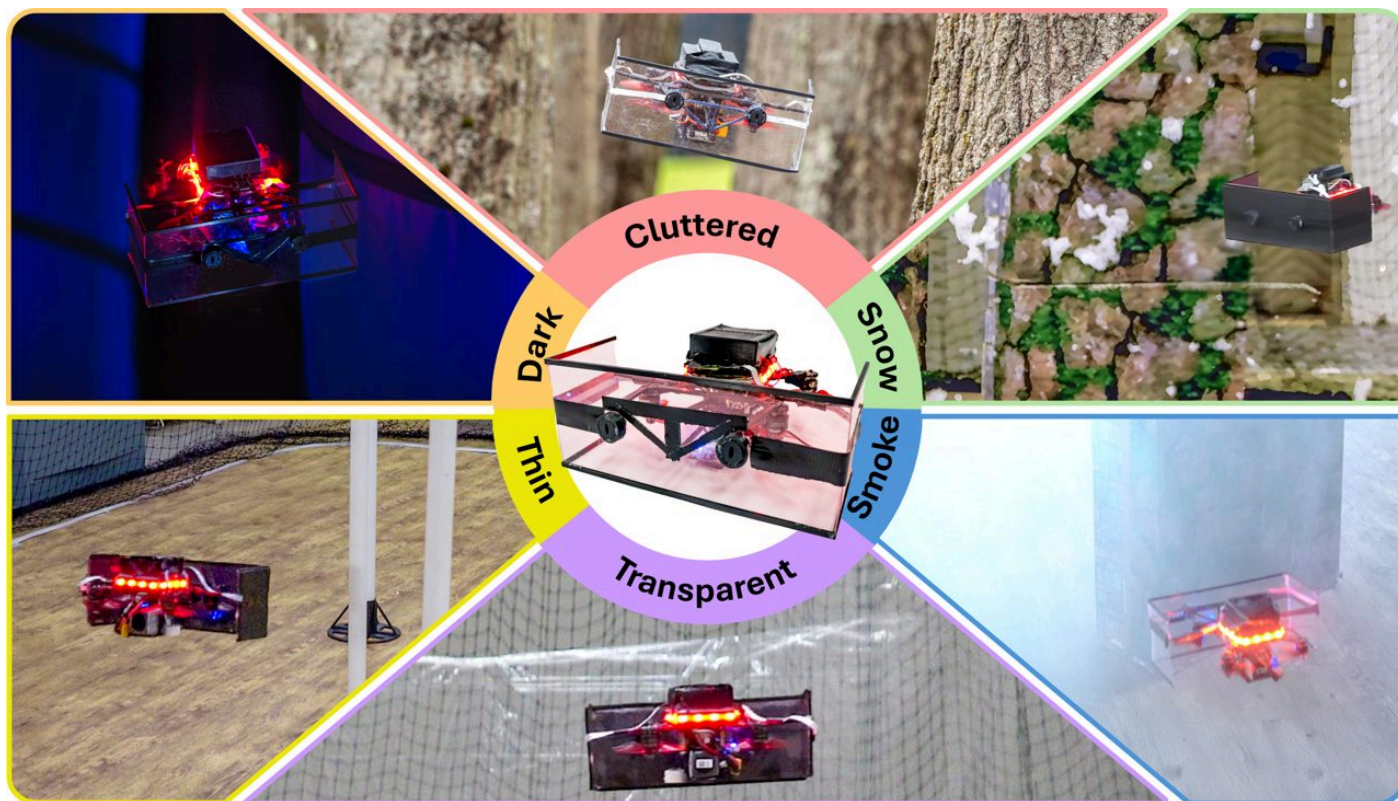
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[Science Robotics 2026] Saranga: milliWatt Ultrasound Navigation on Palm-Sized Drones

Autonomous flight through darkness, smoke, and snow on a 140g drone — milliwatt ultrasound + learned denoising. Science Robotics 2026.



Saranga is the first palm-sized aerial robot capable of autonomous navigation in complete darkness, smoke, and snow — using only onboard milliWatt ultrasound sensing and compute, at just 140g. I employ deep-learning signal processing to suppress noise and estimate depth in real time, achieving 84% success across 10 scenes (200+ trials). Published in Science Robotics (2026).

[Under Review — ICRA 2027] ActiveNav: Learning Active Monocular Flight in Forests

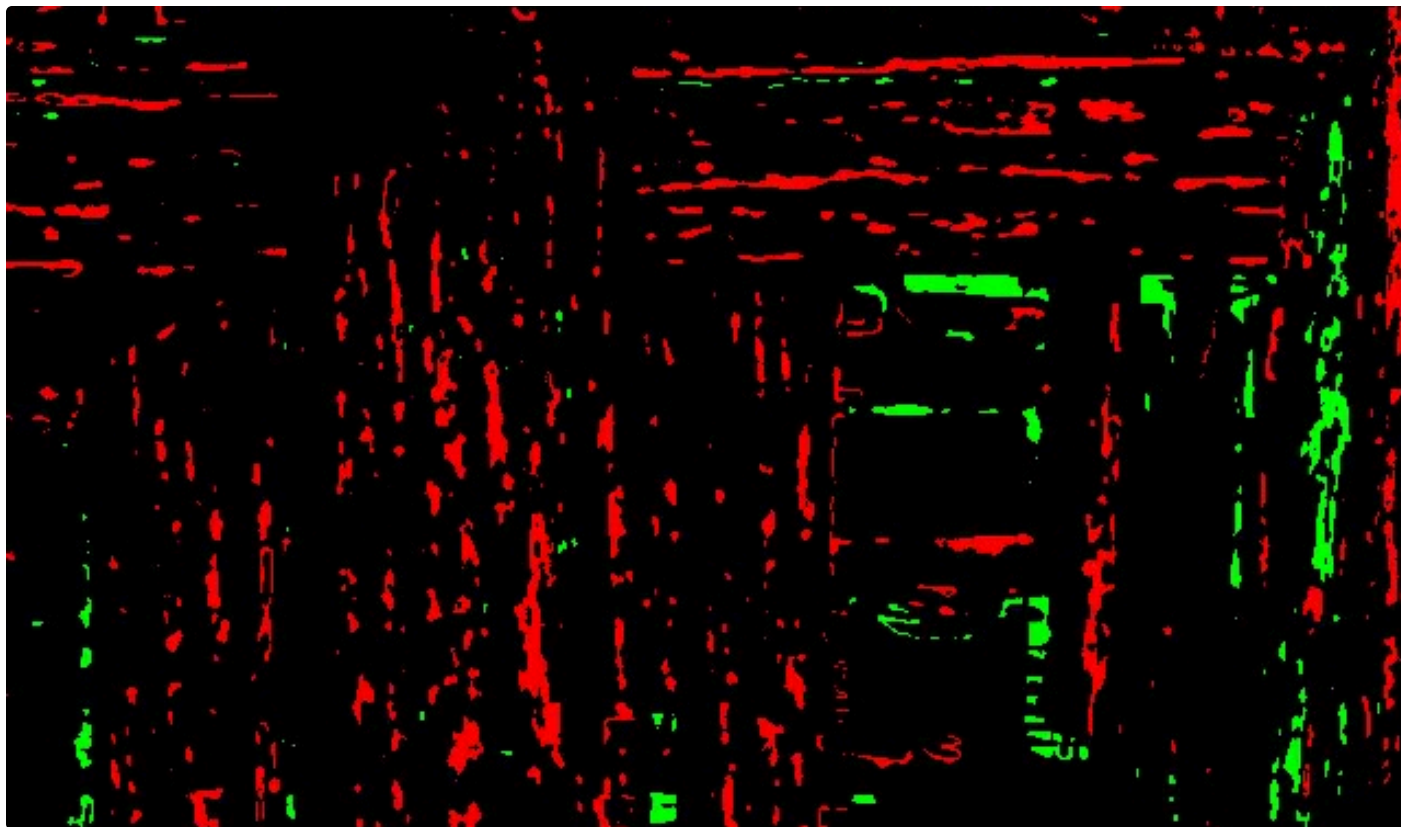
Hierarchical RL that aims the camera mid-flight — 80% success in unseen forests, zero-shot sim2real on a Jetson.



ActiveNav is a novel approach for quadrotor navigation using active perception. With flow and its uncertainty as an input, a hierarchical RL policy actively controls the drone's camera along with its movement. The policy is able to actively point the camera to areas that would improve perception and avoid obstacles. The model was able to generalize zero-shot to the real world, running onboard a Jetson Nano.

[Under Review — RA-L 2026] AttentionSeeker: Passive Attention-Based Aerial Navigation with Events

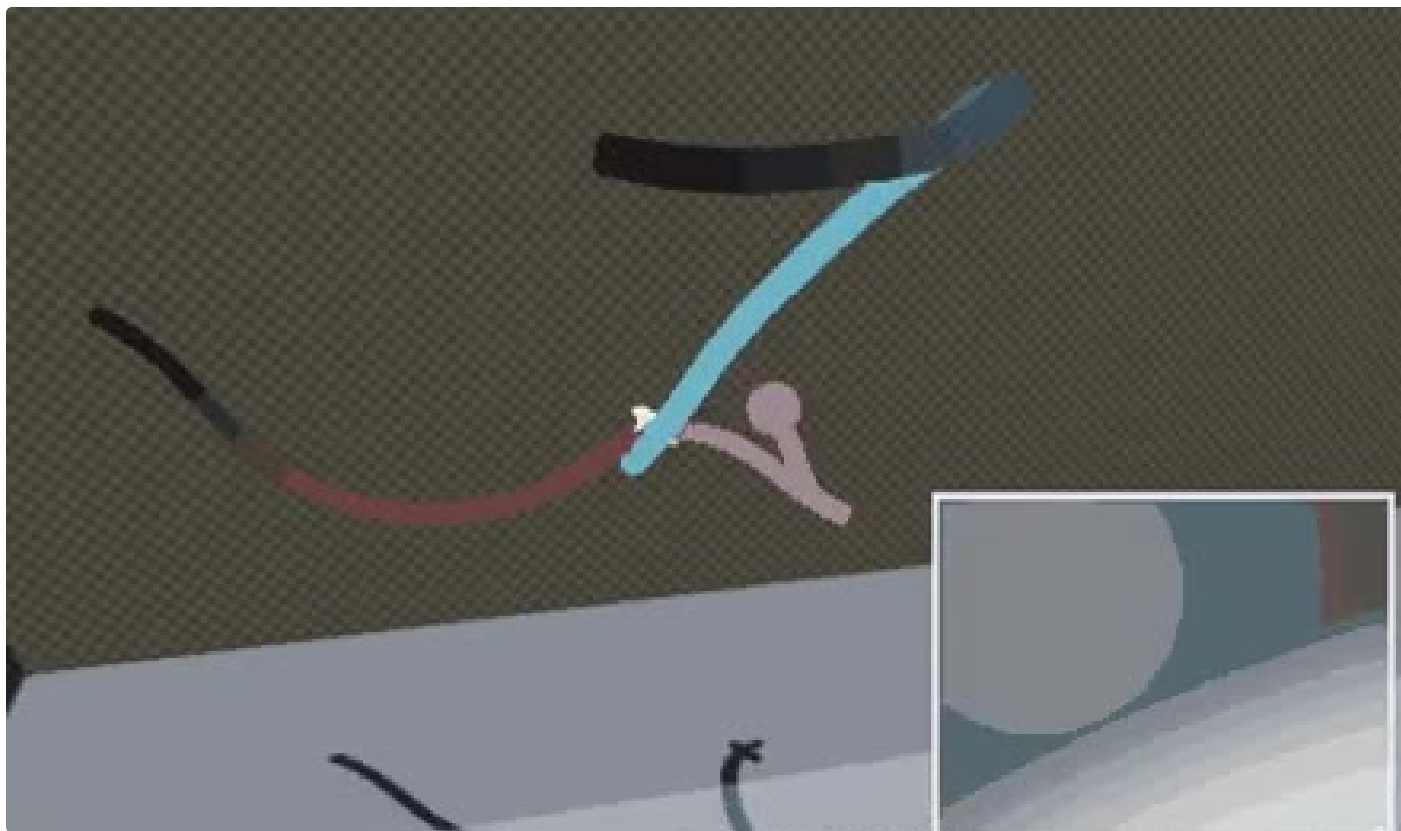
Passive attention from defocus in event streams — high-speed forest flight with no frames and no depth sensor.



AttentionSeeker uses defocus cues in event-camera streams for passive, attention-based aerial navigation. I train reinforcement-learning policies that fly a drone through dense forest at high speed using only events — the GIF above shows one such policy in simulation.

Learning to Chase: Vision-Based Drone Pursuit and Adversarial Self-Play

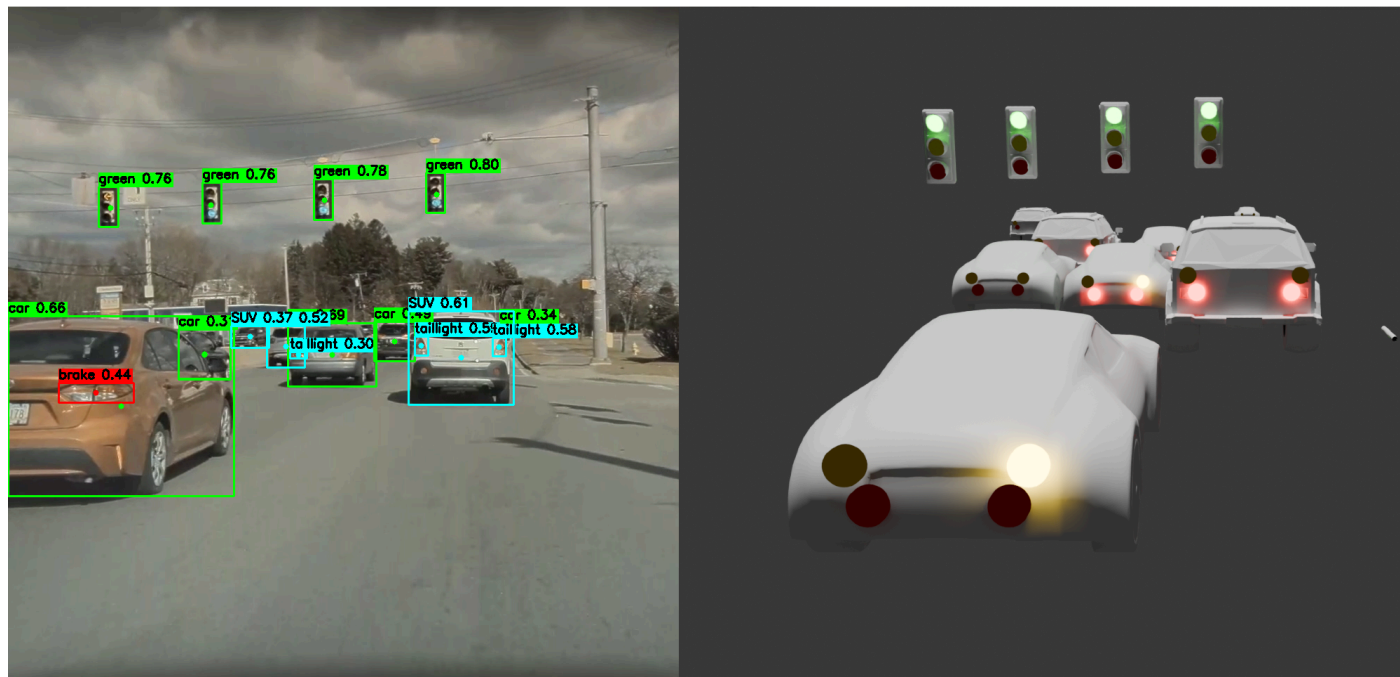
2048 arenas on one GPU: drones that chase a ball from vision alone, and that learn to escape each other.



A GPU-parallel research platform for training quadrotor control policies with reinforcement learning, used to attack two problems: tracking a moving ball from onboard vision alone (the policy never receives the target's position), and 1v1 pursuit–evasion where both drones learn at once. The most valuable findings came from measuring carefully enough to catch my own mistakes — including a noise floor that retracted my headline result.

Einstein Vision: a Full-Self Driving Perception Stack

3D object tracking, lanes, depth, and collision prediction — a full AV perception stack from one camera.



FSD perception stack for a self-driving car, using a variety of models with only a single camera. Pre-trained models were used for object detection, depth estimation, optical flow, and instance segmentation. Objects are tracked in 3D, along with collision prediction, motion detection, and lane detection.